Helmet Detection using YOLOv8n

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1. Problem Statement and Use Case

Motorcycles are one of the most common vehicles on Pakistani roads, and riding without a helmet is a major cause of fatal head injuries. Manual enforcement is slow and cannot cover every road. This project builds an object detection model that looks at an image of a motorcycle rider and detects whether the rider is wearing a helmet or not.

Classes: `helmet`, `no_helmet`

Possible real-world use: an automated traffic-camera system that flags riders without helmets for traffic wardens, or a safety check at the entrance of factories, campuses, and construction sites.

Why this problem: helmet detection has clear, visually distinct classes, which makes consistent annotation realistic for a one-week project, and the use case is directly relevant to road safety in Pakistan.

2. Data Collection

The dataset contains **236 images** in total, collected from two sources:

- **Self-collected images:** photos of motorcycle riders taken by me, with and without helmets. I captured 50 images myself using a laptop and a phone, so the data covers different devices, angles, and lighting.
- **Public images from GitHub:** additional helmet and no-helmet images taken from a public GitHub repository to increase variety. The remaining 186 images came from public GitHub sources. [Add the repository link.]

The images cover different riders, helmet types, backgrounds, and lighting conditions. Duplicate and unusable images were removed before annotation.

3. Annotation

- **Tool:** Roboflow (bounding-box annotation).

- **Method:** part of the images were annotated manually by me in Roboflow, and part were annotated with the help of Roboflow's AI-assisted labeling tools. I annotated 32 images fully manually in Roboflow. The remaining images were labeled with Roboflow's AI-assisted tools. This lower manual-annotation count was discussed with my supervisor and approved by them.
- **Format:** YOLO format (one `.txt` file per image, each line `class x_center y_center width height`, normalized).
- **Annotation rules:** the box covers the rider's head (and helmet if worn); one box per rider; `helmet` is used only when a helmet is actually worn on the head, and `no_helmet` when the head is uncovered or covered only by a cap or cloth.

Data quality check: while evaluating the model, Ultralytics reported one validation label file as corrupt because it mixed polygon (segmentation) rows with bounding-box rows. I converted the polygon rows to bounding boxes (a backup of the original file was kept), cleared the label cache, and re-ran validation. After the fix, all 43 validation images were read with 0 corrupt files. The training and test label files contained no such problem.

4. Dataset Structure and Split

Split	Images	Share
Train	172	73%
Validation	43	18%
Test	21	9%
Total	**236**	100%

The split is close to the required 70/20/10. Images and labels are stored in matching folders:

```
dataset/
  images/{train,val,test}
  labels/{train,val,test}
  data.yaml
```

`data.yaml` defines the dataset path, the three splits, and the two class names.

What each part of `data.yaml` means:

- `path` : the root folder of the dataset on Google Drive; all other paths are relative to it.
- `train` : folder of training images (172 images), used to learn the model weights.
- `val` : folder of validation images (43 images), used during training to monitor progress and pick `best.pt` .
- `test` : folder of test images (21 images), kept aside and used only for the final evaluation.
- `names` : the class list that maps each class id to a name (`0: helmet` , `1: no_helmet` , in the order used in the label files).

5. Model and Training Setup

Setting	Value
Model	YOLOv8n (nano), pretrained COCO weights
Epochs	30
Image size	640
Batch size	8
Framework	Ultralytics 8.4.177, PyTorch 2.11
Hardware	Google Colab, Tesla T4 GPU
Training time	2.2 minutes

YOLOv8n was chosen because it is small and fast, which suits a small dataset and a real-time use case. Starting from pretrained weights lets the model reuse general visual features instead of learning from scratch on only 172 training images.

Training curves: training box, class, and DFL losses all decreased steadily, and the validation losses followed the same trend (after an early spike in validation class loss that settled quickly). Validation precision, recall, mAP50, and mAP50-95 were still rising at epoch 30, which suggests the model had not fully converged and longer training could improve results.

6. Evaluation

Overall results

Split	Images	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Validation	43	0.877	0.705	0.826	0.542
Test	21	0.971	0.701	0.927	0.701

Per-class results

Class	Split	Precision	Recall	mAP@0.5	mAP@0.5:0.95
helmet	Val	0.894	0.720	0.901	0.556
no_helmet	Val	0.861	0.690	0.751	0.527
helmet	Test	1.000	0.652	0.928	0.707
no_helmet	Test	0.941	0.750	0.926	0.694

What the metrics mean:

- **Precision** (about 0.88 on validation, 0.97 on test): when the model predicts a helmet or no-helmet, it is usually correct.
- **Recall** (about 0.70): the model finds roughly 70% of the real objects and misses the rest. This is the weakest metric.
- **mAP@0.5** (0.83 on validation): detection quality at a 50% overlap threshold is good.
- **mAP@0.5:0.95** (0.54 on validation): quality drops under stricter box-overlap requirements, so box placement is only moderately precise.

The test set has only 21 images (28 objects), so the higher test scores should be treated with caution; a few objects can change the numbers noticeably. The validation scores are the more stable estimate.

Best and worst class: on validation, `helmet` performed best (mAP@0.5 0.901) and `no_helmet` was weakest (mAP@0.5 0.751). On the small test set the two classes scored almost the same (0.928 and 0.926). A likely reason for the weaker `no_helmet` class is that it looks more varied (different hair, caps, cloth, and faces), and in low light a face outline can resemble a helmet, which causes wrong-class predictions and weaker boxes.

Confusion matrix (test set, confidence 0.25)

	True helmet	True no_helmet
Predicted helmet	8	0
Predicted no_helmet	1	15
Missed (predicted background)	3	1

The model almost never confuses the two classes with each other (only 1 helmet predicted as no_helmet, and no no_helmet predicted as helmet). Its main weakness is missing helmets (3 of 12) and a few false detections on background regions (8 false `no_helmet` and 2 false `helmet` boxes).

7. Inference on Unseen Images

I tested the final model on **18 new images** that were not part of the train, validation, or test sets, taken at night with different riders, helmet styles, and distances. The `.jfif` files were first converted to `.jpg`. Predictions used a confidence threshold of 0.25, and inference took about 13 ms per image on the T4 GPU.

By visual inspection: **12 images were detected correctly, 5 were missed (no detection), and 1 had the wrong class.** This is a manual visual assessment, not a formal metric.

8. Error Analysis

Image	Actual object	Model prediction	Error type	Possible reason
img_03	helmet (dark full-face)	nothing	False negative	Dark visor blends into night background
img_05	helmet (dark full-face)	nothing	False negative	Dark visor, low light
img_09	helmet (dark full-face)	nothing	False negative	Dark helmet, few similar training examples
img_15	helmet (dark full-face)	nothing	False negative	Dark visor, low light

Image	Actual object	Model prediction	Error type	Possible reason
img_16	no_helmet (extreme close-up)	nothing	False negative	Close-up framing not seen in training
img_04	no_helmet	helmet (0.74)	Wrong class	Low light; face and hair outline resembled a helmet
img_04	background people	no_helmet (0.33)	False positive	Distant people, low confidence threshold
img_02	two riders (correct boxes)	two extra boxes (helmet 0.52, no_helmet 0.35)	False positive	Threshold of 0.25 is too permissive
img_01	helmet	helmet (0.64)	Poor bounding box	Box covers only part of the helmet
img_10	helmet	helmet (0.25)	Weak prediction	Very low confidence; small dataset
img_12	helmet	helmet (0.35)	Weak prediction	Low confidence; small dataset
img_13	no_helmet	no_helmet (0.35)	Weak prediction	Low confidence; small dataset

Error types found: false negatives (5 images), one wrong-class prediction, false positives on background, one poor bounding box, and several low-confidence correct detections. The main causes were poor lighting at night, dark full-face helmets, and a small training set. No clear annotation mistakes were found in the reviewed predictions.

Suggested improvements:

1. Add more images of dark full-face helmets and more low-light and close-up examples for both classes.
2. Train longer (for example 60 epochs) or try YOLOv8s, because the validation metrics were still rising at epoch 30.
3. Raise the inference confidence threshold to about 0.4 to remove weak background boxes.

4. Use brightness and contrast augmentation to make the model more robust at night.

9. Limitations

- Small dataset (236 images), so results may not generalize to all roads, cameras, and weather conditions.
- The test set has only 21 images, which makes test metrics noisy.
- The unseen images were taken at one location at night, with a small number of riders, so they do not represent all real conditions.
- Only two classes were used; the model does not detect the rider (person) or other safety equipment.
- The model was trained for only 30 epochs on a single run; no hyperparameter tuning was done.

10. Future Work

- Collect more images in daylight, rain, crowds, and from traffic-camera angles, with more full-face and dark helmets.
- Train longer (60 or more epochs) and compare against this baseline.
- Try a larger model such as YOLOv8s and stronger augmentation.
- Add a `person` class and number-plate detection for a complete enforcement pipeline.
- Deploy as a simple web or mobile demo.

11. Conclusion

A YOLOv8n model was trained on 236 annotated images for helmet and no-helmet detection. It reached 0.826 mAP@0.5 on validation and 0.927 on test, with high precision and moderate recall. It clearly separates the two classes but misses some dark full-face helmets and low-light cases. With more diverse data and longer training, this approach can become a useful road-safety tool.

12. Project Links

- GitHub repository: [add link]
- Google Colab notebook: [add link]

- LinkedIn post: [add link]